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BONE FIXATION SYSTEM AND METHODS OF USE

Abstract

A bone fixation system comprises a first fastener configured to anchor into a first vertebrae, a second fastener configured to anchor into a second vertebrae, and a set screw to retain the second fastener to the first fastener. The first fastener includes a shaft extending along a longitudinal axis and a head. The shaft includes a first cylindrical body. The head of the first fastener includes a second cylindrical body defining a passageway having an interior surface. The passageway defines a longitudinal axis that is different than the longitudinal axis of the shaft. The first cylindrical body of the shaft and the second cylindrical body of the head are angled relative to each other. A head of the second fastener is configured to engage the interior surface of the passageway of the head of the first fastener. The set screw retains the head of the second fastener in the passageway.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. application Ser. No. 18/166,967, filed Feb. 9, 2023 and which issues as U.S. Pat. No. 12,310,641 on May 27, 2025, which is a continuation application of U.S. application Ser. No. 17/748,816, filed May 19, 2022 and which issued as U.S. Pat. No. 11,576,709 on Feb. 14, 2023, which is a continuation application of U.S. application Ser. No. 17/359,210, filed Jun. 25, 2021 and which issued as U.S. Pat. No. 11,344,349 on May 31, 2022, which is continuation application claiming priority to co-pending international application PCT/US2019/067149, which designates the United States, in the names of Richard G. Fessler et al. entitled “Bone Fixation System and Methods of Use;” and filed on Dec. 18, 2019; which claims priority from and the benefit of the filing date of U.S. Patent Application No. 62/787,571 in the names of Richard Fessler et al entitled “Bone Fixation System and Methods of Use” and filed on Jan. 2, 2019, each of which are hereby incorporated by reference, in their entireties, for all purposes.

FIELD OF INVENTION

[0002] The present invention relates generally to general surgery, orthopedic and neurosurgical implants used for insertion within a patient's vertebrae. More specifically, but not exclusively, the present invention concerns bone fixation systems for implantation into a patient's spine to maintain or re-establish proper spacing and alignment of the spine.

BACKGROUND OF THE INVENTION

[0003] Spinal deformities may result from disease, age or trauma causing destabilization of the spine. To correct destabilization of a patient's spine, posterior fusion device systems may be used. The posterior fusion device systems that are currently available are designed to be applicable to single and multiple level stabilizations using a rod extending between adjacent bone screws for strength. These posterior fusion device systems and the instrumentation used for insertion into a patient's spine are extensive, complicated and expensive and susceptible to loosening of the rods and screws.

SUMMARY OF THE INVENTION

[0004] Aspects of the present invention provide bone fixation systems, including at a single level or multiple levels, and methods that can maintain or re-establish anatomic spacing within a patient's spine.

[0005] According to one aspect of the present invention, provided herein is a bone fixation system, including a first or primary fastener, a second or secondary fastener, and a set screw. The first fastener includes a shaft and an enlarged head. The enlarged head extends from and is angled relative to the shaft. The shaft including a first longitudinal axis. A portion of the shaft of the first fastener is configured to anchor into bone. The enlarged head of the first fastener includes a

passageway having an interior surface. The passageway defines a second longitudinal axis that is dangled with respect to the first longitudinal axis of the shaft. The passageway includes a first end and a second end. The interior surface of the passageway defining a seat. The second fastener includes a shaft extending along a third longitudinal axis and a head. A portion of the shaft of the second fastener is configured to anchor into bone. The head of the second fastener is configured to engage the seat of the first fastener. The engagement of the seat and the head allows for a plurality of angular relationships between the second longitudinal axis relative to the third longitudinal axis. At least a portion of the set screw configured to secure within the passageway of the first fastener. The set screw including a bottom surface that contacts and retains the head of the second fastener against the seat.

[0006] In another aspect, the head of the first fastener and a portion of the second fastener are adapted for residing outside bone.

[0007] These and other objects, features and advantages of this invention will become apparent from the following detailed description of the various aspects and principles of the invention taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention and, together with the detailed description herein, serve to explain the principles of the invention. The drawings are only for purposes of illustrating examples and are not to be construed as limiting the invention. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale.

[0009] In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0010] FIG. 1 illustrates a side view of one embodiment of a bone fixation system, in accordance with one or more aspects of the present invention;

[0011] FIG. 2 illustrates a cross-sectional view of the bone fixation system of FIG. 1, in accordance with one or more aspects of the present invention;

[0012] FIG. 3 illustrates a partial cross-sectional view of the bone fixation system of FIG. 1, in accordance with one or more aspects of the present invention;

[0013] FIG. 4A illustrates a prospective view of one embodiment of a first fastener of the bone fixation system of FIG. 1, in accordance with one or more aspects of the present invention;

[0014] FIG. 4B illustrates a partial cross-sectional view of the first fastener of FIG. 4A, in accordance with one or more aspects of the present invention;

[0015] FIG. 4C illustrates a side view of the first fastener of FIG. 4A, in accordance with one or more aspects of the present invention;

[0016] FIG. 4D illustrates a top view of the first fastener of FIG. 4A, in accordance with one or more aspects of the present invention;

[0017] FIG. 4E illustrates a bottom view of the first fastener of FIG. 4A, in accordance with one or more aspects of the present invention;

[0018] FIG. 5A illustrates a perspective view of one embodiment of a second fastener of the bone fixation system of FIG. 1, in accordance with one or more aspects of the present invention;

[0019] FIG. 5B illustrates a side view of the second fastener of FIG. 5A, in accordance with one or more aspects of the present invention;

[0020] FIG. 5C illustrates a top view of the second fastener of FIG. 5A, in accordance with one or more aspects of the present invention;

[0021] FIG. 5D illustrates a bottom view of the second fastener of FIG. 5A, in accordance with one

or more aspects of the present invention;

[0022] FIG. 5E illustrates a cross-sectional view of the second fastener of FIG. 5A, in accordance with one or more aspects of the present invention;

[0023] FIG. 6 illustrates a prospective view of one embodiment of a drill guide for use with the bone fixation system of FIG. 1, in accordance with one or more aspects of the present invention;

[0024] FIG. 7 illustrates a partial cross-sectional view of the drill guide of FIG. 6 inserted into the first fastener of FIG. 4A, in accordance with one or more aspects of the present invention;

[0025] FIG. 8A illustrates a perspective view of one embodiment of a drill guide for use with the bone fixation system of FIG. 1, in accordance with one or more aspects of the present invention;

[0026] FIG. 8B illustrates a cross-sectional view of the drill guide of FIG. 8A, in accordance with one or more aspects of the present invention;

[0027] FIG. 8C illustrates a side view of the drill guide of FIG. 8A, in accordance with one or more aspects of the present invention;

[0028] FIG. 8D illustrates an end view of the drill guide of FIG. 8A, in accordance with one or more aspects of the present invention;

[0029] FIG. 8E illustrates an opposing end view of the drill guide of FIG. 8A, in accordance with one or more aspects of the present invention;

[0030] FIG. 9 illustrates a side view of another embodiment of a bone fixation system, in accordance with one or more aspects of the present invention;

[0031] FIG. 10 illustrates a partial cross-sectional view of the bone fixation system of FIG. 9, in accordance with one or more aspects of the present invention;

[0032] FIG. 11A illustrates a perspective view of one embodiment of a first fastener of the bone fixation system of FIG. 9, in accordance with one or more aspects of the present invention;

[0033] FIG. 11B illustrates a partial cross-sectional view of the first fastener of FIG. 11A, in accordance with one or more aspects of the present invention;

[0034] FIG. 11C illustrates a side view of the first fastener of FIG. 11A, in accordance with one or more aspects of the present invention;

[0035] FIG. 11D illustrates a top view of the first fastener of FIG. 11A, in accordance with one or more aspects of the present invention;

[0036] FIG. 11E illustrates a bottom view of the first fastener of FIG. 11A, in accordance with one or more aspects of the present invention;

[0037] FIG. 12A illustrates a side view of one embodiment of a second fastener of the bone fixation system of FIG. 9, in accordance with one or more aspects of the present invention;

[0038] FIG. 12B illustrates a cross-sectional view of the second fastener of FIG. 12A, in accordance with one or more aspects of the present invention;

[0039] FIG. 12C illustrates a top view of the second fastener of FIG. 12A, in accordance with one or more aspects of the present invention;

[0040] FIG. 13A illustrates a perspective view of one embodiment of a set screw of the bone fixation system of FIG. 9, in accordance with one or more aspects of the present invention;

[0041] FIG. 13B illustrates a side view of the set screw of FIG. 13A, in accordance with one or more aspects of the present invention;

[0042] FIG. 13C illustrates a top view of the set screw of FIG. 13A, in accordance with one or more aspects of the present invention;

[0043] FIG. 13D illustrates a bottom view of the set screw of FIG. 13A, in accordance with one or more aspects of the present invention;

[0044] FIG. 14 illustrates an exploded side view of another embodiment of a bone fixation system, in accordance with one or more aspects of the present invention;

[0045] FIG. 15 illustrates a cross-sectional view of the bone fixation system of FIG. 14, in accordance with one or more aspects of the present invention;

[0046] FIG. 16A illustrates a partial cross-sectional view of the bone fixation system of FIG. 14, in

accordance with one or more aspects of the present invention;
[0047] FIG. **16B** illustrates another partial cross-sectional view of the bone fixation system of FIG. **14**, in accordance with one or more aspects of the present invention;
[0048] FIG. **17A** illustrates a side view of one embodiment of a first fastener of the bone fixation system of FIG. **14**, in accordance with one or more aspects of the present invention;
[0049] FIG. **17B** illustrates a cross-sectional view of the first fastener of FIG. **17A**, in accordance with one or more aspects of the present invention;
[0050] FIG. **17C** illustrates a top view of the first fastener of FIG. **17A**, in accordance with one or more aspects of the present invention;
[0051] FIG. **17D** illustrates a bottom view of the first fastener of FIG. **17A**, in accordance with one or more aspects of the present invention;
[0052] FIG. **18A** illustrates a perspective view of one embodiment of a second fastener of the bone fixation system of FIG. **14**, in accordance with one or more aspects of the present invention;
[0053] FIG. **18B** illustrates a side view of the second fastener of FIG. **18A**, in accordance with one or more aspects of the present invention;
[0054] FIG. **18C** illustrates a top view of the second fastener of FIG. **18A**, in accordance with one or more aspects of the present invention;
[0055] FIG. **18D** illustrates a bottom view of the second fastener of FIG. **18A**, in accordance with one or more aspects of the present invention;
[0056] FIG. **19A** illustrates a perspective view of one embodiment of a set screw of the bone fixation system of FIG. **14**, in accordance with one or more aspects of the present invention;
[0057] FIG. **19B** illustrates a side view of the set screw of FIG. **19A**, in accordance with one or more aspects of the present invention;
[0058] FIG. **19C** illustrates a top view of the set screw of FIG. **19A**, in accordance with one or more aspects of the present invention;
[0059] FIG. **19D** illustrates a bottom view of the set screw of FIG. **19A**, in accordance with one or more aspects of the present invention;
[0060] FIG. **20** illustrates a perspective view of one embodiment of a drill guide for use with the bone fixation system of FIG. **14**, in accordance with one or more aspects of the present invention;
[0061] FIG. **21** illustrates a partial cross-sectional view of the embodiment of the drill guide of FIG. **20** for use with the bone fixation system of FIG. **14**, in accordance with one or more aspects of the present invention;
[0062] FIG. **22A** illustrates a side view of the drill guide FIG. **20**, in accordance with one or more aspects of the present invention;
[0063] FIG. **22B** illustrates a top view of the drill guide of FIG. **20**, in accordance with one or more aspects of the present invention;
[0064] FIG. **22C** illustrates a bottom view of the drill guide of FIG. **20**, in accordance with one or more aspects of the present invention;
[0065] FIG. **22D** illustrates a cross-sectional view of the drill guide of FIG. **20**, in accordance with one or more aspects of the present invention;
[0066] FIG. **23** illustrates a perspective view of another embodiment of a bone fixation system, in accordance with one or more aspects of the present invention;
[0067] FIG. **24** illustrates a partial perspective view of the bone fixation system shown in FIG. **23**, in accordance with one or more aspects of the present invention;
[0068] FIG. **25** illustrates an exploded view of the bone fixation system shown in FIG. **23**, in accordance with one or more aspects of the present invention;
[0069] FIG. **26A** illustrates one perspective view of one embodiment of a head for use with the bone fixation system of FIG. **23**, in accordance with one or more aspects of the present invention;
[0070] FIG. **26B** illustrates a side view of the head shown of FIG. **26A**, in accordance with one or more aspects of the present invention;

[0071] FIG. 26C illustrates another perspective view of the head shown of FIG. 26A, in accordance with one or more aspects of the present invention;

[0072] FIG. 26D illustrates a cross-sectional view of the head shown of FIG. 26A, in accordance with one or more aspects of the present invention;

[0073] FIG. 27A illustrates a perspective view of one embodiment of a first fastener of the bone fixation system of FIG. 23, in accordance with one or more aspects of the present invention;

[0074] FIG. 27B illustrates a side view of the first fastener of FIG. 27A, in accordance with one or more aspects of the present invention;

[0075] FIG. 27C illustrates a top view of the first fastener of FIG. 27A, in accordance with one or more aspects of the present invention;

[0076] FIG. 27D illustrates a bottom view of the first fastener of FIG. 27A, in accordance with one or more aspects of the present invention;

[0077] FIG. 28A illustrates a side view of one embodiment of a second fastener of the bone fixation system of FIG. 23, in accordance with one or more aspects of the present invention;

[0078] FIG. 28B illustrates a top view of the second fastener of FIG. 28A, in accordance with one or more aspects of the present invention;

[0079] FIG. 28C illustrates a bottom view of the second fastener of FIG. 28A, in accordance with one or more aspects of the present invention;

[0080] FIG. 29A illustrates a perspective view of one embodiment of a spacer of the bone fixation system of FIGS. 23-25, in accordance with one or more aspects of the present invention;

[0081] FIG. 29B illustrates a side view of the spacer of FIG. 29A, in accordance with one or more aspects of the present invention;

[0082] FIG. 29C illustrates a top view of the spacer of FIG. 29A, in accordance with one or more aspects of the present invention;

[0083] FIG. 29D illustrates a bottom view of the spacer of FIG. 29A, in accordance with one or more aspects of the present invention;

[0084] FIG. 30A illustrates a perspective view of one embodiment of a set screw of the bone fixation system of FIG. 23-25, in accordance with one or more aspects of the present invention;

[0085] FIG. 30B illustrates a side view of the set screw of FIG. 30A, in accordance with one or more aspects of the present invention;

[0086] FIG. 30C illustrates a top view of the set screw of FIG. 30A, in accordance with one or more aspects of the present invention; and

[0087] FIG. 30D illustrates a bottom view of the set screw of FIG. 30A, in accordance with one or more aspects of the present invention.

DETAILED DESCRIPTION FOR CARRYING OUT THE INVENTION

[0088] Generally stated, disclosed herein are different embodiments and examples of a bone fixation system. The embodiments and examples described herein of a bone fixation system constructed in accordance with one or more aspects of the present invention avoid the need for using a rod to connect adjacent pedicle screws typically used in back surgery. Without use of a connecting rod, there are less parts involved and less movable parts and less chance of movement of the fusion system or loosening of its components after surgery with a bone fixation system constructed in accordance with one or more principles or aspects of the present invention. A bone fixation system constructed in accordance with one or more principles or aspects of the present invention also does not rely on a connecting rod for strength. Further, surgical methods for inserting the bone fixation systems are discussed.

[0089] In this detailed description and the following claims, the words proximal, distal, anterior, posterior, medial, lateral, superior, inferior, cephalad and caudally are defined by their standard usage for indicating a particular part of a bone or implant according to the relative disposition of the natural bone or directional terms of reference. For example, “proximal” means the portion of an implant nearest the insertion instrument, while “distal” indicates the portion of the implant farthest

from the insertion instrument. As for directional terms, “anterior” is a direction towards the front side of the body, “posterior” means a direction towards the back side of the body, “medial” means towards the midline of the body, “lateral” is a direction towards the sides or away from the midline of the body, “superior” means a direction above and “inferior” means a direction below another object or structure, “cephalad” means a direction toward the head and “caudally” means a direction toward the inferior part of the body.

[0090] In general, the embodiments and examples described herein of a bone fixation system constructed in accordance with one or more principles or aspects of the present invention include a primary screw or fastener and a secondary screw or fastener. The primary fastener or fastener includes a shaft and a head angled relative to the shaft. In one example, the secondary fastener is directly coupled to the primary fastener by, for example, mating threads or an interference fit. In alternative embodiments, a set screw is used to retain the second fastener within the head of the primary fastener. In other examples, the head of the primary fastener provides a seat for the second fastener to enable variable angulations of the secondary fastener relative to the primary fastener.

[0091] In general, the head of the primary fastener is shaped and angled relative to the shaft to enable insertion of the secondary fastener through it and into an adjacent pedicle, thereby eliminating the need for a rod or plate. The shape of the head of the primary fastener provides a low profile and less trauma to soft tissues when inserted, for example, via minimally invasive techniques. In one example, the head of the primary fastener is enlarged with respect to the shaft of the primary fastener. For example, the cross-sectional width of the enlarged head, taken normal to the longitudinal axis of the head, is greater than the cross-sectional width of the shaft, taken normal to the longitudinal axis of the shaft. The head may include a body (e.g. cylindrical) that is angled relative to the shaft of the primary fastener. The position of the head of the primary fastener ensures closest proximity of the head to bone, thereby maximizing strength of the system. The angulation of the head relative to the shaft of the primary fastener enables appropriate direction of the secondary fastener to engage a second vertebra or pedicle, thus eliminating the need for a rod or plate. By minimizing the number of parts, the risk of failure through, for example, loosening of set screws to tulip heads in conventional bone fixation systems is reduced.

[0092] In general, the head of the primary fastener and a portion of the second fastener are configured to reside outside bone to allow for trajectory of the secondary fastener to engage adjacent pedicle or use at multiple adjacent vertebrae levels. The head of the primary fastener is incorporated partially into the longitudinal axis of the shaft of the primary fastener to minimize the overall size of the head, thereby decreasing soft tissue trauma on insertion and long-term irritation of adjacent muscle. The angulated head of the primary fastener enables, for example, in line insertion of a primary fastener screwdriver, and insertion of a secondary fastener and screwdriver through the same external entrance of the primary fastener during insertion.

[0093] The bone fixation system described herein may also include a tapering or a gradually increasing thickness of the shaft of the primary fastener as it approaches the head. This tapering or increased thickness at a region of greatest stress on the primary fastener reduces failure through fracture of the primary fastener and enhances the fastener to bone interface thereby providing enhanced stability by increasing compressive load.

[0094] The combination of the primary fastener and secondary fastener described by the examples herein enables insertion of the bone fixation system via either an MIS or OPEN technique without the need for additional instrumentation. The system design relies on angulation of two fasteners relative to each other for stability and minimizes the need for additional instrumentation or components, such as, for example, rods and multiple screw lengths and thicknesses. The design of the bone fixation system described herein combines the resistance to “pull out” imparted by the threads of the fasteners with that created by the angulation of the fasteners relative to each other, thereby making pull out virtually impossible.

[0095] Referring to the drawings, wherein like reference numerals are used to indicate like or

analogous components throughout the several views, and with particular reference to FIGS. **1-8E**, there is illustrated an exemplary embodiment of a bone fixation system **100**. The bone fixation system **100** may include a first or primary fastener **110** for anchoring into, for example, a first bone or vertebrae and a second or secondary fastener **150** for anchoring into, for example, a second bone or vertebrae.

[0096] With continued reference to FIGS. **4A-4E**, first fastener **110** may include a shaft **120**, a head **130**, and a longitudinal axis **121**. Shaft **120** may include a neck **112** connecting shaft **120** to the head **130** and one or more bone engagement mechanisms **122** to facilitate a gripping engagement of first fastener **110** to bone. Shaft **120** also includes an outer surface **114**. In one embodiment, outer surface **114** may taper radially outward from longitudinal axis **112** in the portion of neck **112** proximate head **130**.

[0097] In one example illustrated in FIG. **4A**, shaft **120** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external thread, for example, is a single lead thread that extends from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0098] As shown in FIGS. **4A-4E**, head **130** includes a cylindrical body **131**. In one embodiment, the diameter of cylindrical body **131** is greater than the diameter of shaft **120**, when measured normal to their respective longitudinal axes. When connected to neck **112**, head **130** appears like a halo over shaft **120**. Head **130** may form a receiving portion, such as, for example, a passageway **132** having a longitudinal axis **134**. Passageway **132** may be in the form of, for example, a through hole or, alternatively, a slot formed between two arms. Passageway **132** defines an interior cavity or space having an inner surface **136** extending from a first end **138** to a second end **140**. In one embodiment, first end **138** and second end **140** extend beyond the outer surface **114** of shaft **120**. Passageway **132** may be sized and configured to receive and allow pass through of at least a portion of second fastener **150** or other spinal connection element for anchoring the system **100** to, for example, a second bone or vertebrae. Passageway **132** may also be sized and configured to retain or secure at least a portion of head **180** of second fastener **150**.

[0099] In one embodiment illustrated in FIG. **4B**, interior surface **136** of passageway **132** may include, for example, a threaded portion **142** that extends along at least a portion of interior surface **136** from first end **138**. Passageway **132** may also include a seat **144** that may extend along at least another portion of interior surface **136** towards second end **140**. In one example, seat **144** may be formed by interior surface **136** tapering inward towards longitudinal axis **134** as it approaches second end **140**. A tool engagement opening **124** (see FIG. **4D**) may be formed within passageway **132** and extending into neck **114** of shaft **120** for engagement with, for example, a screwdriver for inserting first fastener **110** into the bone. Tool engagement opening **124** may align with a cannulated bore extending through first fastener **110**.

[0100] As illustrated in FIGS. **5A-5E**, second fastener **150** may include a longitudinal axis **162**, a proximal end **164**, a distal end **166**, a shaft **170**, a head **180** and a neck **168** connecting shaft **170** to head **180**. Shaft **170** may include one or more bone engagement mechanisms **172** to facilitate a gripping engagement of second fastener to bone. Shaft **170** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external thread is a single lead thread that extend from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0101] As shown in FIGS. **5A, 5C and 5E**, head **180** may include a top surface **182**, a first side portion **184** and a second side portion **186**. Top surface **182** may include a tool engagement opening

extending into head **180**. First side portion **184** may include external threads **188** extending along at least a portion of the exterior surface of first side portion **184** from proximal end **164**. External threads **188** are configured to engage threaded portion **142** of interior surface **136** of passageway **132** of first fastener **110**. In one example illustrated in FIGS. **3** and **5E**, second side portion **186** of head **180** may include an exterior surface **190** that tapers inward towards longitudinal axis. The tapered exterior surface **190** is configured to engage and mate with seat **144** of passageway **132** and allow first fastener **110** to direct second fastener **150** seamlessly through passageway **132**, while also preventing second fastener **150** from migrating entirely through passageway **132** of first fastener **110**.

[0102] The system **100** may be implanted by first preparing a patient's vertebrae for insertion of the bone fixation system **100**. In one example, the preparation may be done using minimally invasive procedures. Next, a surgeon obtains a first fastener **110** and a second fastener **150**. First fastener **110** is anchored or inserted into a desired bone such as, for example, a first vertebrae. In one example, a surgeon drives first fastener **110** into the first vertebrae using a screw driver engaged with tool engagement opening **124**. Next, second fastener **150** is inserted through passageway **132** and anchored into another desired bone, such as, for example, a second or adjacent vertebrae. Second fastener **150** is inserted into passageway **132** until threads **188** of first side portion **184** mate or engage threaded portion **142** on surface **136** in passageway **132** of first fastener **110**. In securing second fastener **150** to first fastener **110**, second fastener **150** is screwed into head **130** until tapered exterior surface **190** of second fastener **150** engages seat **144** in passageway **132** of first fastener **110**. In the system **100** illustrated in FIGS. **1-8E**, second fastener **150** is orientated in a fixed angled relationship relative to first fastener **110**. This fixed angle is created by engagement of threads **188** of first side portion **184** and tapered surface **190** of second side portion threads **142** of head **180** of second fastener **150** with threaded portion **142** and seat **144** of passageway **132** of head **130** of first fastener **110**. When properly installed, the entire head **130** and a portion of shaft **120** of first fastener **110** and/or shaft **170** of second fastener **150** are positioned and reside outside the bone(s) in which first fastener **110** and second fastener **150** are inserted. By positioning head **130** outside the vertebrae, bone fixation system **100** is configured to aid in the fusion of multi-levels of vertebrae.

[0103] Prior to insertion of second fastener **150** through first fastener **110** and into a second desired bone, a drill guide may be used to pre-drill a hole in the bone. FIGS. **6-8E** illustrate one embodiment of a drill guide that may be used in system **100**. As illustrated in FIG. **8A**, drill guide **192** may include a hollow elongated shaft **193** having a first end **194**, a second end **195**, a handle portion **196** proximate second end **195** and a head **198** at first end **194**. Head **198** includes an exterior surface **199**. Exterior surface **199** may be configured, sized and shaped to engage threaded portion **142** and seat **144** of interior surface **136** of head **130** of first fastener **110**. As illustrated in FIG. **7**, head **198** of drill guide **192** is received by and inserted into passageway **132**. The engagement of threads on exterior surface **199** of drill guide **192** and threaded portion **142** on interior surface **136** provide stability of drill guide **192** in relation to first fastener **110** for pre-drilling the hole for second fastener **150**.

[0104] Another embodiment of a bone fixation system **200** is shown in FIGS. **9-13D**. Bone fixation system **200** may include a first or primary fastener **210** for anchoring into, for example, a first vertebrae, a second or secondary fastener **250** for anchoring into, for example, a second vertebrae, and a set screw or locking cap **214**.

[0105] With continued reference to FIGS. **11A-11E**, first fastener **210** may include a shaft **220**, a head **230** and a longitudinal axis **221**. Shaft **220** may include a neck **212** connecting shaft **220** to the head **230**. Shaft **220** also includes an outer surface **214**. A portion of the outer surface **214** of shaft **220** may include one or more bone engagement mechanisms **222** to facilitate a gripping engagement of first bone anchor to bone. In one example illustrated in FIG. **11A**, shaft **220** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-

threaded. In the illustrated embodiment, the external thread is a single lead thread that extends from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0106] As shown in FIG. 11B, head **230** includes a cylindrical body **231**. In one embodiment, the diameter of cylindrical body **231** is greater than the diameter of shaft **220** extending to outer surface **214**. When connected to neck **212**, head **230** appears like a halo over shaft **220**. Head **230** may form a receiving portion, such as, for example, a passageway **232** having a longitudinal axis **234**. Passageway **232** may be in the form of, for example, a through hole or, alternatively, a slot formed between two arms. Passageway **232** defines an interior cavity or space having an inner surface **236** extending from a first end **238** to a second end **240**. In one embodiment, first end **238** and second end **240** extend beyond the outer surface **214** of shaft **220**. Passageway **232** may be sized and configured to receive and allow pass through of a second fastener **250** or other spinal connection element for anchoring the system **200** to, for example, a second bone or vertebrae. Passageway **232** is also configured to retain or secure set screw **214** and at least a portion of head **280** of second fastener **250**.

[0107] In one embodiment illustrated in FIG. 11B, interior surface **236** of passageway **232** may include, for example, a threaded portion **242** that extends along at least a portion of interior surface **236** from first end **238**. Passageway **232** may also include a seat **244** that may extend along at least another portion of interior surface **236** towards second end **240**. In one example, seat **244** may be formed by interior surface **236** tapering inward towards longitudinal axis **234** as it approaches second end **240**.

[0108] As illustrated in FIGS. 12A-12C, second fastener **250** may include a longitudinal axis **262**, a proximal end **264**, a distal end **266**, a shaft **270**, and a head **280**. Shaft **270** may include one or more bone engagement mechanisms **272** to facilitate a gripping engagement of second fastener to bone. Shaft **270** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external thread, for example, is a single lead thread that extends from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0109] As shown in FIG. 12A, head **280** may include a top surface **282** and a side portion **284**. Top surface **282** may include a tool engagement opening (see FIG. 12C) extending into head **280**. Side portion **284** of head **280** may include an exterior surface **290** that tapers inward towards longitudinal axis **262**. The tapered exterior surface **290** is configured to engage and mate with seat **244** of passageway **232**.

[0110] As shown in FIGS. 13A-13D, set screw **214** may include a top surface **215**, side surface **217** and a bottom surface **219**. Set screw **214** may also include a tool engagement opening **216** extending into set screw **214** from top surface **215** toward bottom surface **219**. In addition, set screw **214** may include threads **218** on side surface **217** extending, for example, from top surface **215** to bottom surface **219**, or along a portion thereof. Bottom surface **219** may be sized and shaped to engage at least a portion of head **280** of second fastener **250**.

[0111] The system **200** may be implanted by first preparing a patient's vertebrae for insertion of the bone fixation system **200**. Next, the first fastener **210** is screwed or inserted into the bone, e.g. vertebrae. In inserting first fastener **210** into the bone, a surgeon may use a screw driver that engages a tool engagement opening **235** (see FIG. 11D) formed within passageway **230**. Next, second fastener **250** is inserted through head **230** of first fastener **210** and into a second desired bone until exterior tapered surface **290** engages seat **244** of passageway **232** of head **230** of first fastener **210**. When properly installed, the entire head **230** and a portion of shaft **120** of first

fastener **210** and/or shaft **270** of second fastener **250** are positioned and reside outside the bone(s) in which first fastener **210** and second fastener **250** are inserted. By positioning head **230** outside the vertebrae, bone fixation system **200** is configured to aid in the fusion of multi-levels of vertebrae.

[0112] Prior to insertion of second fastener **250** into first fastener **210**, drill guide **192** may be used to pre-drill a hole in the bone. Next, set screw **214** may be inserted or screwed into passageway **232** to secure second fastener **250** in the desired position. Finally, the patient's incision may be closed. In the system **210** illustrated in FIGS. 9-13D, second fastener **250** is orientated in a fixed angled relationship relative to first fastener **210**. This fixed angle is created by engagement of tapered surface **290** of side portion **284** of head **280** of second fastener **250** with seat **244** of passageway **232** of first fastener **210**.

[0113] Another embodiment of a bone fixation system **300** is shown in FIGS. 14-22D. Bone fixation system **300** may include a first or primary fastener **310** for anchoring into, for example, a first vertebrae, a second or second fastener **350** for anchoring into, for example, a second vertebrae, and a set screw or locking cap **314**. In one example, first fastener **310** may be cannulated with a threaded halo head, second fastener **350** may be cannulated with an articulating spherical head, and set screw **314** may be cannulated and include a bottom surface for capturing and securing the spherical head of second fastener **350**.

[0114] As shown in FIGS. 17A-17D, first fastener **310** may include a cannulated shaft **320**, a head **330** and a longitudinal axis **321**. Shaft **320** may include a neck **312** connecting shaft **320** to the head **330** and an outer surface **313**. In one example, as illustrated in FIGS. 17A and 17B, a portion of outer surface **313** of neck **312** may taper radially outward relative to longitudinal axis **321** to provide additional thickness to neck **312** of shaft **320** as it approaches head **330**. This taper or increased thickness provides, for example, enhancement of the fastener to bone interface thereby providing enhanced stability by increasing compressive load and reduces failure through fracture of the fastener.

[0115] A portion of outer surface **313** of shaft **320** may include one or more bone engagement mechanisms **322** to facilitate a gripping engagement of first bone anchor to bone. In one example illustrated in FIG. 17A, shaft **320** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external threads, for example, may be a single lead thread that extend from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0116] As illustrated in FIG. 17B, head **330** includes a cylindrical body **331**. In one embodiment, the diameter of cylindrical body **331** is greater than the diameter of shaft **320** extending to outer surface **313**. Head **330** may form a receiving portion, such as, for example, a passageway **332** having a longitudinal axis **334**. Passageway **332** may be in the form of, for example, a through hole or, alternatively, a slot formed between two arms. Passageway **332** defines an interior cavity or space having an inner surface **336** extending from a first end **338** to a second end **340**. In one embodiment, first end **338** and second end **340** extend beyond the outer surface **313** of shaft **320**. Passageway **332** may be sized and configured to receive and allow pass through of a second fastener **350** or other spinal connection element for anchoring the system **300** to, for example, a second bone or vertebrae. Passageway **332** is also configured to retain or secure set screw **314** and at least a portion of head **380** of second fastener **350**.

[0117] In one embodiment illustrated in FIG. 17B, interior surface **336** of passageway **323** may include, for example, a threaded portion **342** that extends along at least a portion of interior surface **336** from first end **338**. Passageway **332** may also include a seat **344** that may extend along at least another portion of interior surface **336** towards second end **340**. In one example, seat **344** may be shaped to correspond and mate with the shape of a portion of head **380** of second fastener **350**.

[0118] Passageway **332** is in communication with the cannulated tube or passageway extending through shaft **320** of first fastener **310**. In one example, passageway **323** and the cannulated tube or passageway extending through shaft **320** are configured to receive a K-wire to assist in inserting first fastener **310** into the bone or vertebrae at a proper location and position. A tool engagement opening **324** such as, for example, a cannulated hex receiver hole, is formed in interior surface **336** and into shaft **320** between passageway **323** and the cannulated tube **326**. Tool engagement opening **324** receives a hex driver or screwdriver and assists in inserting or screwing first fastener **310** into the bone.

[0119] As illustrated in FIGS. **18A-18D**, second fastener **350** may include a longitudinal axis **362**, a proximal end **364**, a distal end **366**, a cannulated shaft **370**, a head **380** and a neck **368** connecting shaft **370** to head **380**. The outer surface of shaft **370** may include one or more bone engagement mechanisms **372** to facilitate a gripping engagement of second fastener to bone. Shaft **370** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external thread is a single lead thread that extend from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0120] As illustrated in FIG. **18A**, head **380** may include a top surface **382** and a side portion **384**. Top surface **382** may include a tool engagement opening **383** extending into head **380**. Side portion **384** of head **380** may include an exterior surface **390** shaped like, for example, a polyaxial screw head in the form of, for example, a bulbous head, hemispherical, or partially spherical head. Exterior surface **390** may be shaped to lie in or mate with seat **344** formed in interior surface **336** of passageway **332** of head **330** of first fastener **310** near second end **340**. Seat **344** preferably has a shape that matches or corresponds to the shape of a portion of the polyaxial side portion **384** and allows, for example, pivoting, spinning and rotation of second fastener **350** relative to longitudinal axis **334** of head **330** of first fastener **310**. This configuration allows a range of motion along several different axes, e.g. multi-directional movement or rotation or angulation, of second fastener **350** relative to longitudinal axis **334**. For example, a round head **380** of second fastener **350** mating with a mating concave-shaped seat **344** enables a variation of angulation to accommodate pedicle screw insertion in variable anatomy. Such a round head **380** transfers the variable angulation from the tulip head of a traditional pedicle screws to the secondary fastener **350**.

[0121] After second fastener **350** is positioned into bone in a selected direction and orientation relative to first fastener **310**, set screw **314**, as illustrated in FIGS. **19A-19D**, may lock the orientation of second fastener **350** relative to first fastener **310**. Set screw **314** may have any suitable size, configuration and means for securing polyaxial screw head **380** to seat **344** in a selected orientation relative to first fastener **310**. In one embodiment shown in FIGS. **19A-19D**, set screw **314** may include a top surface **315**, side surface **317** and a bottom surface **319**. Set screw **314** may also include a tool engagement opening **316** extending into set screw **314** from top surface **315** toward bottom surface **319**. In one example, set screw **314** may be cannulated, but in other examples may not. In addition, set screw **314** may include threads **318** on side surface **317** extending, for example, from top surface **315** to bottom surface **319**, or along a portion thereof. As shown in FIGS. **16A** and **16B**, bottom surface **319** is sized and shaped to retain head **380** with the aid of seat **344**.

[0122] The concave contour of bottom surface **319** of set screw **380** is configured to match or correspond to head **380** of second fastener **350** in order to, for example, maximize surface contact and, therefore, maximize the rigidity of the system. In one embodiment, bottom surface **219** may include a surface treatment, such as, for example, surface etching, which engages and/or interacts with head **380** of second fastener **350** to provide, for example, enhanced rigidity through a friction fit that prevents shifting of second fastener **350** once secured. In one example, head **380** may also

include a surface treatment or similar surface etching that engages and/or interacts with bottom surface **319** of set screw **314**. When properly installed, the entire head **330** and a portion of shaft **320** of first fastener **310** and/or shaft **370** of second fastener **350** are positioned and reside outside the bone(s) in which first fastener **310** and second fastener **350** are inserted. By positioning head **330** outside the vertebrae, bone fixation system **300** is configured to aid in the fusion of multi-levels of vertebrae. For example, head **330** positioned outside the vertebrae allows for a wider range of angular relationship between first fastener **310** and second fastener **350** for fastening to adjacent vertebrae or vertebrae spaced farther up or down the spine from each other.

[0123] Prior to insertion of second fastener **350** through first fastener **310** and into a second desired bone, an angle guide may be used to select the appropriate angle of second fastener **350** with respect to first fastener **310**. FIGS. **20-22D** illustrate one embodiment of an angle guide that may be used in system **300**. As illustrated in FIGS. **22A-22D**, angle guide **392** may include a hollow elongated shaft **393** having a first end **394**, a second end **395**, a handle portion **396** proximate second end **395** and a head **398** at first end **394**. Head **398** includes an exterior surface **399**. Exterior surface **399** may be configured, sized and shaped to engage at least a portion of seat **344** of interior surface **336** of head **330** of first fastener **310**. As illustrated in FIG. **21**, head **398** of angle guide **392** is received by and inserted into passageway **332** until exterior surface **399** engages seat **344** to then allow a drill to be used to pre-drill a hole prior to insertion of second fastener **350**. Exterior surface **399** may include a mating bulbous configuration to seat **344**, matching, in one example, bulbous portion of exterior surface **390** of head **380** of second fastener **350**, to allow angular positioning of angle guide **392** within passageway **332** of head **330**. In this example, a surgeon may be able to position angle guide **392** at a desired angle relative to longitudinal axis **334** of head **330**.

[0124] In one example, system **300** may be inserted into the spine of a patient. System **300** may be implanted by first preparing a patient's vertebrae for insertion of the bone fixation system **300**. With a patient in the prone position, a surgeon may make an incision approximately **2** cm to midline. Next, the proper location and orientation of first or primary fastener **310** is determined. Using an A/P fluoroscopy, the cephalad pedicle is identified. A Jamshidi needle may then be placed through the soft tissues so that the tip is in the middle of the cephalad pedicle. Next, the Jamshidi needle is then positioned so that a "bullseye" image is seen on A/P fluoroscopy, guaranteeing a direct trajectory into the pedicle. A K-wire is then placed through the Jamshidi approximately **1** cm into the facet bony structure. The use of a K-wire technique via Jamshidis and cannulated screws ensures correct trajectories prior to insertion of the first and second fasteners without the need for image guidance, thereby improving safety while decreasing cost.

[0125] Fluoroscopy is then placed into the lateral position to conform that the trajectory is appropriate. The Jamshidi is removed and the K-wire is advanced through the pedicle into the vertebral body. Next, over the K-wire, a "pilot" hole is created using a tap. A cannulated screw driver tool is engaged with engagement opening **324** of first fastener **310**. First fastener **310** (with the cannulated screw driver tool) is then placed over the K-wire and first fastener **310** is inserted or screwed into the cephalad pedicle. The K-wire is then removed and the position of first or primary fastener **310** is confirmed with lateral fluoroscopy. After first fastener **310** is properly located, the cannulated screw driver tool is removed.

[0126] Next, the area is prepared for insertion of second or secondary fastener **350**. First, head **398** of angle guide **392** is inserted into passageway **330** of head **330** of first fastener **310**. A Jamshidi needle is then placed through angle guide **392**. Using lateral fluoroscopy, the proper and appropriate trajectory for second fastener **350** to intersect the caudal pedicle is determined. When the proper location is determined, the surgeon gently taps the Jamshidi needle into the bone to secure the appropriate position. Next, a K-wire is placed through the Jamshidi needle and the trajectory is confirmed though lateral and A/P Fluoroscopy. Under lateral fluoroscopy, the K-wire is advanced into the caudal pedicle, and the Jamshidi needle and angle guide **392** are removed. After

angle guide **392** is removed, a pilot hole is drilled using a cannulated drill bit over the K-wire. Then, a surgeon will confirm position on lateral and A/P Fluoroscopy before removing the K-wire. [0127] Next, a cannulated screw driver tool engages tool engagement opening **383** of second fastener **350**. Second fastener **350** is then advanced over the K-wire to the desired depth until head **380** fully seats onto seat **344** formed in interior surface **336** of passageway **332** of head **330** of first fastener **310**. The placement of second fastener **350** is then verified using lateral fluoroscopy. Next, a surgeon uses a cannulated screw driver tool to engage tool engagement opening **316** of set screw **314**. Set screw **314** (with the cannulated screw driver tool) is placed over the same K-wire and inserted or screwed into head **330** of first fastener **310** until bottom surface **319** of set screw **314** contacts head **380** of first fastener **310** and secures head **380** against seat **344**. Next, the final placement of system **300** is confirmed using A/P and lateral fluoroscopy and then the K-wire is removed. Finally, the patient's incision may be closed.

[0128] In the system **300** illustrated in FIGS. **19-19D**, second fastener **350** may be of fixed orientation or be orientated in various angles relative to first fastener **110** depending on the designed embodiment. The specific angle configuration of second fastener **350** relative to first fastener **310** is locked into place by set screw **314** within passageway **332**. In one example, system **300** may have the ability to angulate second fastener **350** approximately twelve degrees in any direction on spherical or bulbous head **380** relative to axis **334** of first fastener **310** that is necessary when a Jamshidi needle or K-wire is inserted through second fastener **350**. Without a needle or K-wire insert through second fastener **350**, system **300** may have the ability to angulate second fastener **350** up to approximately forty degrees relative to axis **334** of first fastener **310**. In one example, the angle configuration of longitudinal axis **321** of first fastener **310** relative to longitudinal axis **362** of second fastener **350** ranges between approximately thirty-three degrees to approximately sixty-seven degrees. In another example, when longitudinal axis **362** of second fastener **350** is aligned with axis **334** of head **340**, the angle between longitudinal axis **321** of first fastener **310** relative to longitudinal axis **362** of second fastener **350** is approximately fifty degrees.

[0129] Another embodiment of a bone fixation system **400** is shown in FIGS. **23-30D**. Bone fixation system **400** may include a first fastener **410** for anchoring into, for example, a first vertebrae, a second fastener **450** for anchoring into, for example, a second vertebrae, a head **430**, a spacer **402** and a set screw or locking cap **412**.

[0130] As shown in FIGS. **27A-27D**, first fastener **410** may include a shaft **420**, a head **425** and a longitudinal axis **421**. Shaft **420** may include a neck **414** connecting shaft **420** to the head **425**. Shaft **420** includes an outer surface **414**. Shaft **420** may include one or more bone engagement mechanisms **422** on at least a portion of outer surface **414** to facilitate a gripping engagement of first bone anchor to bone. In one example illustrated in FIG. **27A**, shaft **420** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external thread is a single lead thread that extend from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism. Head **425** may include a top surface **426** and a side portion **427**. Top surface **426** may include a tool engagement opening extending into head **424**. Side portion **427** of head **425** may include an exterior surface shaped like, for example, a polyaxial screw head in the form of, for example, a bulbous head, hemispherical, or partially spherical head.

[0131] As illustrated in FIGS. **28A-28C**, second fastener **450** may include a proximal end **464**, a distal end **466**, a shaft **470**, and a head **480**. Shaft **470** may include one or more bone engagement mechanisms **472** to facilitate a gripping engagement of second fastener to bone. Shaft **470** may be, for example, threaded along its entire length, threaded along only a portion of the length, or non-threaded. In the illustrated embodiment, the external thread is a single lead thread that extend from a distal tip of the shaft to the proximal head portion. Other suitable bone engagement mechanisms

may include, but are not limited to, one or more annular ridges, multiple threads, dual lead threads, variable pitched threads, longitudinal splines or other geometries, and/or any conventional bone engagement mechanism.

[0132] As illustrated in FIGS. 26A-26D, head **430** includes a cylindrical body **431**. In one embodiment, the diameter of body **431** is greater than the diameter of shaft **420** extending to outer surface **414**. Head **430** may include a top end **432**, a bottom end **434**, a base **436** extending from bottom end **434**, and a first arm **438** and a second arm **440** extending upward from base **436** to top end **432**. As illustrated in FIG. 26A, head **430** may form receiving portions for receiving both first fastener **410** and second fastener **450**. Head **430** may define a first passageway **442** along axis **444** for receiving first fastener **410** and a second passageway **446** along axis **448** for receiving second fastener **450**. In one example, axis **444** of first passageway **442** may be perpendicular to axis **448** of second passageway **446**. In one example, first passageway **442** may be formed as a through hole extending from first end **432** to second end **434**. In one embodiment, first end **432** and second end **434** extend beyond outer surface **414** of shaft **420**. Second passageway **446** may be formed as a U-shaped slot or other suitable opening for receiving second fastener **450** and retaining, at least, a portion of head **480** of second fastener **450**.

[0133] In one example shown in FIG. 26D, U-shaped slot of a second passageway **446** includes a U-shaped interior surface **433** formed by the inner surfaces of first arm **438** and second arm **440** and a top surface **438** of base **436**. This U-shaped slot may be sized and configured to receive head **480** of second fastener **450**. Interior surface **433** may include, for example, a threaded portion **431** that extends along at least a portion of interior surface **433** from first end **432**. Interior surface **433** may also include, for example, a non-threaded portion **437** that extends along at least a portion of interior surface **433** from first end threaded portion **431** to and including top surface **438** of base **436**.

[0134] As illustrated in FIG. 26D, passageway **442** may also include a seat **449** that may extend along at least another portion of interior surface **433** within base **436** towards second end **434**. In one example, seat **449** may be shaped to correspond and mate with the shape of a portion of outer surface **427** of head **425** of first fastener **410**. Exterior surface **427** may be shaped to lie in or mate with seat **449** formed in interior surface of passageway **442** near second end **434**. Seat **449** preferably has a shape that matches the shape of the exterior surface **427** and allows, for example, pivoting, spinning and rotation of first fastener **410** relative to axis **444**. This configuration allows a range of motion along several different axes, e.g. multi-directional movement or rotation, of first fastener **410** relative to longitudinal axis **444**. In this embodiment, the angulation achieved by system **400** with respect to second fastener **450** relative to first fastener **410** may be similar to that achieved by the example provided for system **300**.

[0135] As illustrated in FIGS. 30A-30D, set screw or locking cap **412** may include a top surface **413**, side surface **414** and a bottom surface **415**. Set screw **412** may also include a tool engagement opening **416** extending into set screw **412** from top surface **213** toward bottom surface **415**. In addition, set screw **412** may include threads **417** on side surface **414** extending, for example, from top surface **413** to bottom surface **415**, or along a portion thereof. Bottom surface **415** may be sized and shaped to engage at least a portion of the side surface of head **480** of second fastener **450**. Threads **417** are sized and configured to engage threaded portion **431** that extends along at least a portion of interior surface **433** from first end **432** of head **430**.

[0136] As illustrated in FIGS. 29A-29D, spacer **402** includes a top surface **403**, a bottom surface **404** and a side surface **405**. Spacer **402** is sized and configured to be disposed between top surface **426** of first fastener **410** and the side surface of head **480** of second fastener **450**.

[0137] The system **400** may be implanted by first preparing a patient's vertebrae for insertion of the bone fixation system **400**. Next, a surgeon obtains a first fastener **410** and head **430**. First fastener **410** is inserted through first passageway **442** until head **425** of first fastener **410** engages seat **449** within base **436** of head **430**. First fastener **410** is then inserted into bone in a selected direction and

orientation. Next, spacer **402** is positioned onto of top surface **426** of first fastener **410**. Then, second fastener **450** is inserted through second passageway **446** and positioned into bone in a selected direction and orientation relative to first fastener **410**. Second fastener **450** is locked into place by set screw **412**. Head **480** of second fastener **450** is locked between bottom surface **415** of set screw **412** and top surface **403** of spacer **402** as set screw **412** is screwed into passageway **442**. As set screw **412** is screwed down onto the side surface of head **480**, first fastener **410** is also being set into place through the force translated through the contact of spacer **402** with the side surface of head **480** of second fastener **450** and top surface **426**. In the system **400** illustrated in FIGS. **23-30D**, first fastener **410** may be orientated in various angles relative to second fastener **450**. The specific angle configuration of first fastener **410** relative to second fastener **450** is locked into place set screw **412** within passageway head **430**.

[0138] Similar to the prior embodiments, a drill guide may be used to pre-drill holes in the bone for both first fastener **410** and second fastener **450**. Alternative constructions and configurations of a drill guide may also be used in accordance with the principles of the present invention.

[0139] In each of the embodiments described herein, first and second fasteners, drill guides and other components may include through holes to accommodate, for example, K-wires, drills and Jamshidi needles to assist with positioning and orientation during surgery. For example, first fastener **110** and second fastener **150** include through holes along their longitudinal axis to accommodate, for example, K-wires. In addition, alternative methods may be employed to install the various systems described herein. For example, a surgical robot or an image guidance system may be used to perform or aid in performing all or portions of the various steps required to properly install a system constructed in accordance with one or more aspects within a patient.

[0140] Although the fasteners shown and described through the present description, such as fasteners **110**, **210**, **310** and **410** and **150**, **250**, **350** and **450** are shown as being straight, it is also contemplated that one or more of these fasteners may have curved shafts that may be driven into the patient's vertebrae rather than screwed, or the shafts of these fasteners may also be slightly curved.

[0141] As may be recognized by those of ordinary skill in the art based on the teachings herein, numerous changes and modifications may be made to the above-described and other embodiments of the present invention without departing from the scope of the invention. The fasteners, elongate members, and other components of the devices and/or systems as disclosed in the specification, including the accompanying abstract and drawings, may be replaced by alternative component(s) or feature(s), such as those disclosed in another embodiment, which serve the same, equivalent or similar purpose as known by those skilled in the art to achieve the same, equivalent or similar results by such alternative component(s) or feature(s) to provide a similar function for the intended purpose. In addition, the devices and systems may include more or fewer components or features than the embodiments as described and illustrated herein. For example, the components and features of FIGS. **1-30D** may all be used interchangeably and in alternative combinations as would be modified or altered by one of skill in the art. Accordingly, this detailed description of the currently-preferred embodiments is to be taken as illustrative, as opposed to limiting the invention.

[0142] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has”, and “having”), “include” (and any form of include, such as “includes” and “including”), and “contain” (and any form of contain, such as “contains” and “containing”) are open-ended linking verbs. As a result, a method or device that “comprises,” “has,” “includes,” or “contains” one or more steps or elements possesses those one or more steps or elements, but is not limited to possessing only those one or more steps or elements. Likewise, a step of a method or an element of a device that “comprises,”

“has,” “includes,” or “contains” one or more features possesses those one or more features, but is not limited to possessing only those one or more features. Furthermore, a device or structure that is configured in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0143] The invention has been described with reference to the preferred embodiments. It will be understood that the architectural and operational embodiments described herein are exemplary of a plurality of possible arrangements to provide the same general features, characteristics, and general system operation. Modifications and alterations will occur to others upon a reading and understanding of the preceding detailed description. It is intended that the invention be construed as including all such modifications and alterations.

Claims

1. A bone fixation device, comprising: a first fastener comprising a first threaded shaft for engaging bone tissue and a first fastener head extending from a proximal portion of the first threaded shaft; a second fastener comprising a second threaded shaft for engaging bone tissue and a second fastener head extending from a proximal portion of the second threaded shaft; and a head, wherein the head comprises: a cylindrical body; a first passageway extending through the cylindrical body from a first end to a second end forming a seat in a distal end of the head for receiving a portion of the first fastener; and a second passageway extending through the cylindrical body to form a U-shaped slot for receiving a portion of the second fastener.
2. The bone fixation device of claim 1, wherein an axis of the first passageway extends through the head perpendicular to an axis of the second passageway.
3. The bone fixation device of claim 1, wherein the head further comprises: a base extending from a bottom end of the head toward a top end; a first arm extending upward from the base to the top end; and a second arm extending upward from the base to the top end on a side opposite the first arm.
4. The bone fixation device of claim 3, wherein an inner surface of the first arm and an inner surface of the second arm form an interior surface of the U-shaped slot.
5. The bone fixation device of claim 4, wherein the interior surface of the U-shaped slot comprises a threaded portion extending along at least a portion of the interior surface.
6. The bone fixation device of claim 5, wherein the seat extends along at least a portion of the interior surface within the head and is configured to receive the first fastener head.
7. The bone fixation device of claim 5, wherein the seat is shaped to correspond with the shape of an outer surface of the first fastener head.
8. The bone fixation device of claim 7, wherein the first fastener pivots within the seat of the head relative to a longitudinal axis of the first fastener.
9. The bone fixation device of claim 1, wherein the seat comprises a concave-shaped surface configured to receive a convex rounded portion of the second fastener head.
10. The bone fixation device of claim 1, wherein the head is configured to receive a set screw in a threaded engagement with the interior surface to contact the head of the multi-axial bone fastener.
11. The bone fixation device of claim 10, wherein the head is configured to receive a spacer positioned between an interior surface of the head and an interior surface of the set screw.
12. The bone fixation device of claim 1, wherein a diameter of the cylindrical body is greater than a diameter of the first threaded shaft of the first fastener.
13. The bone fixation system of claim 10, wherein the set screw is configured to fix an orientation of the first fastener head of the first fastener against the seat of the first passageway of the head.
14. The bone fixation system of claim 10, wherein a bottom surface of the set screw is a concave surface that mates with a portion of the second fastener head of the second fastener.
15. The bone fixation system of claim 1, wherein the second fastener head has a cylindrical shape.

16. A method of implanting a bone fixation system, comprising: obtaining the bone fixation system with a first fastener, a second fastener, and a head; preparing a patient's vertebrae for insertion of the bone fixation system; inserting the first fastener through a first passageway in the head until a first fastener head of the first fastener engages a seat within a base of the head; inserting the coupled first fastener and head into the patient's bone; inserting the second fastener through a second passageway in the head and into the patient's bone in a direction and orientation relative to the first fastener

17. The method of claim 16, further comprising: inserting a spacer into the first passageway of the head until positioned on top of the first fastener head of the first fastener.

18. The method of claim 17, further comprising: inserting a set screw into the first passageway of the head to lock the second fastener and the first fastener into place.

19. The method of claim 16, further comprising: inserting a set screw into the first passageway of the head to lock the second fastener into place.

20. The method of claim 16, further comprising: inserting a set screw partially into the first passageway of the head; adjusting an angle between the first fastener and the second fastener to a desired orientation; and screwing the set screw fully into the first passageway of the head to lock the second fastener and the first fastener into the desired orientation.
